



RMC.BestFit 2.0 Verification Report

Analytical oracles, independent package comparisons, published benchmarks, recovery studies, and coverage evidence

U.S. ARMY CORPS OF ENGINEERS, RISK MANAGEMENT CENTER

Draft for external peer review - not for public release

BestFit 4304fb39f8e1 / Numerics 90a63a46394d

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This external-review draft is generated from the publication manifest. A claim is verified only when its test design, oracle or recovery basis, acceptance rule, and result are stated.

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CHAPTER 01

Report Documentation

Document control

Field	Value
Title	RMC.BestFit 2.0 Verification Report
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Abstract

This report presents numerical verification evidence for RMC.BestFit 2.0, a Bayesian-first statistical analysis framework used in flood-frequency and water-resources engineering. Verification is based on analytical calculations, independently implemented numerical oracles, external scientific packages, published examples, parameter-recovery experiments, and simulation-coverage experiments. The report states the software configuration, test data, oracle provenance, acceptance criteria, observed results, and supported conclusion for each claim. It follows the application order: time-series data; input data; distribution fitting; univariate and Bulletin 17C analyses; point-process, competing-risk, mixture, and composite analyses; bivariate and coincident-frequency analyses; rating curves; AR, MA, ARIMA, and ARIMAX analyses; and spatial extremes. Deterministic unit and regression tests are reported as software-quality controls but are not substituted for scientific verification. Evidence boundaries are stated explicitly where the current checkpoint does not support a broader claim.

Review and release notice

This document is a technical draft supplied for independent scientific and engineering review. A report number, named reviewers, approval signatures, and final distribution statement will be assigned after comment disposition. Numerical conclusions apply only to the software and dependency checkpoints identified above and to the parameter regions, data designs, and acceptance rules stated in this report.

Revision record

Revision	Date	Description
External-review draft	28 August 2026	Reorganized the verification record around test design, oracle provenance, numerical results, and conclusions; refreshed the documented software checkpoint and publication controls.

Suggested citation

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Executive Summary

Purpose and scope

The purpose of this report is to establish which scientific calculations in RMC.BestFit 2.0 have been compared with evidence independent of the production implementation. The report does not attempt to prove correctness for every possible model, data set, prior, or tail probability. It documents bounded claims whose test design, reference result, and acceptance rule can be inspected and reproduced.

The report follows the application's project-tree order: time-series data; input data; distribution fitting; univariate, Bulletin 17C, point-process, competing-risk, mixture, and composite analyses; bivariate and coincident-frequency analyses; rating-curve analysis; AR, MA, ARIMA, and ARIMAX analyses; and spatial extremes. Maximum-likelihood, maximum-a-posteriori, generalized-method-of-moments, Bayesian estimation, model comparison, and convergence diagnostics are shared foundations.

Verification conclusion

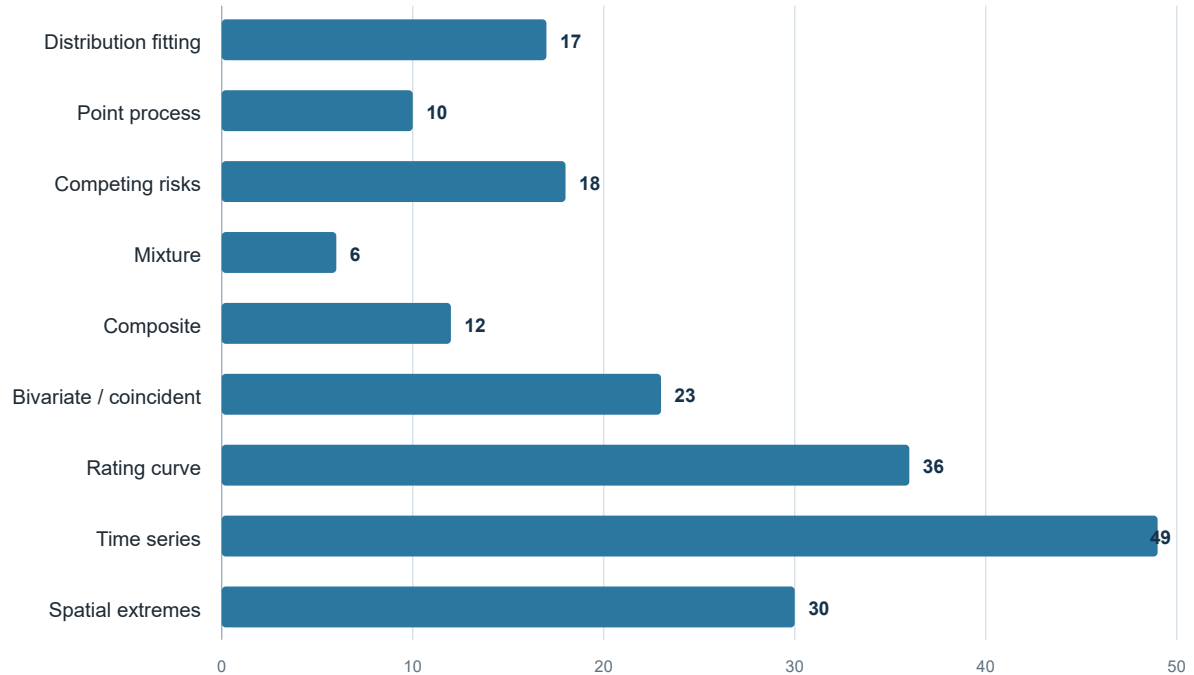
At the stated software checkpoint, the independently supported cells summarized below satisfy their declared acceptance criteria. The strongest evidence is exact analytical or external-package parity. Recovery evidence is conditional on its generating model, sample size, seed, estimator configuration, and acceptance rule. Simulation coverage is claimed only where a predeclared repetition design and Monte Carlo acceptance interval were executed for the stated checkpoint.

Time-series-data and input-data persistence, validation, and processing contracts passed the fast regression gate. Those two collection-level results control the handoff to the scientific models but are not counted as independent numerical verification.

Capability	Verification basis	Result at the accepted checkpoint
Fifteen univariate families and multi-candidate fitting	SciPy, R <code>lmomco</code> , analytical identities, and independent fitting oracles	17 of 17 fitting cells passed
Model comparison and convergence diagnostics	R <code>loo</code> , R <code>posterior</code> , R <code>gmm</code> , R <code>bbmle</code> , and analytical covariance results	All reported oracle comparisons passed
Bulletin 17C worked examples	Published example parameters and PeakFQ plotting positions	7 of 7 parameter examples and 3 of 3 plotting-position comparisons passed
Bulletin 17C refit reliability	Ordinary and pivotal resampling of seven published examples	14 of 14 cells; 13,000 finite outputs; zero retries, substitutions, or exceptions
Point-process models	Analytical Poisson/GPA calculations, likelihood identities, and recovery	10 of 10 cells passed
Competing risks	Analytical dependence identities and generating-model recovery	4 of 4 analytical, 10 of 10 MLE, and 4 of 4 supported Bayesian cells passed
Finite mixtures	Independent Numerics parity and Bayesian generation-recovery	6 of 6 cells passed
Composite analyses	Closed forms, R <code>mistr</code> , Gaussian orthants, and complete Cartesian posterior enumeration	12 of 12 reported cells passed
Bivariate and coincident frequency	Independent copula optimum, Bayesian recovery, and closed-form Normal sums	23 of 23 reported cells passed
Rating curves	SciPy likelihood and optimum, analytical continuity, and curve recovery	36 of 36 reported cells passed
Time-series models	Independent recurrence/transform oracles and generating-model recovery	12 oracle, 8 independent recovery, and 29 supporting recovery cells passed
Spatial extremes	R <code>mvtnorm</code> , conditional-GP and haversine oracles, model recovery, and analysis-level checks	30 of 30 reported cells passed

Passed verification cells by analysis area

Accepted oracle and recovery evidence at the documented checkpoint



Counts summarize evidence cells and should not be interpreted as code coverage or equal scientific weight.

Figure 1. Passed cells in the principal recovery and oracle matrices. Counts are evidence cells, not a measure of scientific importance or code coverage. Model-estimation diagnostics and Bulletin 17C reliability are tabulated separately because their evidence units differ.

Interpretation

A passing cell supports the claim stated for that cell. It does not establish universal accuracy outside the tested support, asymptotic regime, sample size, dependence structure, or prior configuration. In particular:

- The six difficult competing-risk Bayesian designs not included in the supported matrix are research-grade identifiability and convergence cases and are not claimed as verified.
- Current Bulletin 17C Cohn-value and broad coverage claims are excluded because the corresponding numerical comparisons were not executed at this checkpoint.
- Bivariate copula fitting conditions on fixed fitted marginals; joint marginal-copula posterior estimation is not claimed.
- Spatial weighting is a heuristic influence weighting and not a composite pairwise likelihood or effective-sample-size correction.
- Generic prior-predictive sampling is not a joint-prior sampler when a model contains coupled prior factors beyond independently sampled parameter priors.

These boundaries are limitations of the supported claim set, not evidence that the verified calculations failed.

CHAPTER 03

System Under Test and Verification Methodology

System under test

All results in this draft are tied to the following reproducible configuration.

Component	Checkpoint
RMC.BestFit	Version 2.0.0; commit <code>4304fb39f8e162cdb746083108042df88e153afc</code>
RMC.Numerics source used for peer review	Commit <code>90a63a46394db9ef95e72b0fcba943408110636</code>
Published package compatibility baseline	RMC.Numerics 2.1.4
Runtime	.NET on Windows, 64-bit process
Verification execution	One fully qualified method per test process, with a uniquely named TRX record
Bayesian workhorse	DEMCzs production defaults unless a test states otherwise
Default reproducibility seed	12345 unless a fixture declares another seed

The source checkpoint and package baseline answer different questions. The source checkpoint identifies the numerical code used for the peer-review evidence. The package baseline identifies the published dependency version against which public compatibility and serialized payloads were checked.

Terminology

Verification asks whether the software implements the specified equations and algorithms correctly. **Validation** asks whether a model is an adequate representation of the physical process for a particular application. This report primarily provides verification. Recovery and coverage studies contribute limited validation evidence for their declared simulation designs, but they do not validate a model for every field application.

Evidence hierarchy

Evidence is classified before results are interpreted:

1. **Analytical oracle.** A closed-form value, identity, derivative, recurrence, or hand calculation independent of the production path.
2. **Independent numerical implementation.** A separately written R or Python calculation with its generator, environment, source inputs, and hash recorded.
3. **External scientific package.** A calculation from packages such as R `loo`, `posterior`, `gmm`, `bbmle`, `mvtnorm`, `copula`, or Python SciPy.
4. **Published result.** A table, worked example, or standard with sufficient parameterization detail to reproduce the comparison.
5. **Recovery experiment.** Data are generated from known parameters and refitted; acceptance is based on parameter, curve, quantile, and diagnostic criteria.
6. **Coverage experiment.** Repeated simulated data sets test whether nominal confidence or credible intervals contain the generating quantity at the declared rate.

An estimator that merely completes, converges, returns finite values, or agrees with another production path is not independently verified. Constructor, validation, state, cache, serialization, exception, and fixed-calculation tests remain in the fast regression projects and are not counted as scientific verification cells.

Deterministic comparisons

For a scalar reference value r and software value s , absolute and relative errors are

$$e_{\text{abs}} = |s - r|, \quad e_{\text{rel}} = \frac{|s - r|}{\max(|r|, e_0)},$$

where e_0 is the stated scale floor. A cell passes when its declared absolute or relative error is no greater than the predeclared tolerance. Vector and matrix comparisons apply the same rule elementwise unless the test states a norm or specialized statistic.

Recovery experiments

Recovery tests predeclare the generating model, retained sample size, seed, fitted coordinates, and response quantity. Maximum-likelihood recovery compares the fitted optimum with the independent optimum or generating parameter. Bayesian recovery ordinarily requires:

- the generating parameter or curve to lie inside the stated central posterior interval;
- rank-normalized $\hat{R} < 1.1$ for monitored coordinates;
- effective sample size greater than the declared minimum, commonly 100; and
- a point or curve estimate within the declared relative tolerance.

Recovery is evaluated on the scientifically interpretable quantity whenever parameter non-identifiability makes individual coordinates misleading. Mixture weights and rating-curve segments are examples where ordered or curve-based assertions are important.

Coverage experiments

For B independent repetitions and indicator I_b that the interval contains the truth, empirical coverage is

$$\hat{C} = \frac{1}{B} \sum_{b=1}^B I_b.$$

The Monte Carlo standard error at nominal coverage C_0 is

$$\text{MCSE}(\hat{C}) = \sqrt{\frac{C_0(1 - C_0)}{B}}.$$

A coverage claim is reported only when the repetition count, nominal target, completion rule, and Monte Carlo acceptance interval were declared before execution. Re-enabled but unexecuted assertions are not reported as results.

Bayesian diagnostics

Rank-normalized split $\hat{R}$, bulk effective sample size, and tail effective sample size are compared with R `posterior`. PSIS-LOO values, smoothed importance weights, Pareto k , and importance-sampling effective sample size are compared with R `loo`. A numerically equal aggregate alone is insufficient; the pointwise decomposition and diagnostic vectors are also compared.

Artifact and execution controls

Independent oracle artifacts are committed beneath `verification/data/` with generation metadata and SHA-256 entries in the data manifest. Recovery fixtures record their generating recipe, sample size, seed, parameterization, and acceptance thresholds. Long-running methods are executed separately to prevent fixed result logs or shared test hosts from coupling otherwise independent evidence cells. The complete verification project is not used as a publication command; the report is supported by the explicit method-level evidence set described in its chapters and internal claim-evidence ledger.

Controlled publication refresh

The following exact-method refresh was completed on 24 August 2026 at the controlled BestFit and Numerics commits in Table 1. It is a reproducibility check on selected cells from the larger evidence matrices reported in the analysis chapters; it does not replace or enlarge those claim sets.

Refresh group	Test design	Result
Analytical, external-package, published-source, and independently implemented oracle methods	One exact method and one TRX record per guarded invocation	55 of 55 passed
Representative MLE, Bayesian, likelihood, and posterior-propagation recovery methods	Point process through spatial extremes, using the predeclared seeds and acceptance rules	15 of 15 passed
Total	No production algorithm, tolerance, sampler setting, seed, or reference contract changed	70 of 70 passed

CHAPTER 04

Time-Series Data Verification Boundary

Test objective

The software's first project collection stores dated values, interval and unit metadata, source provenance, and diagnostic plots. The publication question at this boundary is whether the project preserves the data supplied to the scientific model; it is not whether an external data provider or a hydrologic record is correct.

Test and result

Fast UI tests exercise manual construction, validation and property transitions, project persistence and copy behavior, and HEC-DSS import contracts. These tests passed in the publication regression gate. They are deterministic software-contract evidence and are not counted as independent numerical verification.

Independent recurrence, transform, likelihood, generation, and recovery tests consume controlled dated series later in this report. Their passing results establish that the model layer interprets the tested chronology and intervals as stated, but they do not validate arbitrary downloaded records.

Evidence boundary

The report makes no claim about network availability, provider revisions, station quality, missing-value treatment chosen by an analyst, or the suitability of a series for a specific risk assessment. Those checks precede model verification.

CHAPTER 05

Input-Data Verification Boundary

Test objective

The second project collection converts manual, agency, block-series, or peaks-over-threshold inputs into the `DataFrame` consumed by fitting and univariate analyses. The verified boundary is the preservation of observation type, chronology, index, censoring/threshold semantics, and processing configuration.

Test and result

Fast UI and core tests exercise collection persistence, dirty-state propagation, validation, processing state, plotting-position contracts, exact/interval/threshold/uncertain series construction, and deterministic block/peak calculations. These tests passed in the publication regression gate. They establish data-contract behavior but are not counted as external numerical or recovery evidence.

Downstream chapters describe the independent tests that consume committed frames: distribution fits and functions, Bulletin 17C worked examples, point-process mixed-observation likelihoods, rating-curve aligned pairs, and spatial missing-row likelihoods. All reported downstream results use explicit fixtures and acceptance rules.

Evidence boundary

Input processing does not prove stationarity, independence, record completeness, perception-threshold correctness, or application suitability. The analyst must validate those assumptions before relying on a downstream frequency model.

CHAPTER 06

Distribution Fitting, Univariate Analysis, and Bulletin 17C

Verification objectives

This chapter verifies that supported univariate families use the documented parameterizations, likelihood measure, support, distribution functions, and fitted optima; that fitting-analysis criteria and weights are computed from those fitted results; and that the specialized Bulletin 17C generalized-method-of-moments implementation reproduces the published worked-example parameters.

Distribution-family oracles

The family tests use deterministic data committed with the report. Each test compares the fitted parameter vector, maximized data log likelihood, representative CDF values, and representative quantiles with an independent implementation. SciPy 1.17.1 supplies thirteen overlapping families. R 4.4.3 with `lmomco` 2.5.7 supplies Generalized Logistic and Generalized Normal. Closed-form calculations supplement Normal, Log10-Normal, Exponential, and Logistic, and analytical branch identities supplement Kappa Four.

Family	Primary independent reference	Quantities compared	Result
Normal	Closed-form MLE and SciPy	Parameters, likelihood, CDF, quantiles	Passed
Log10-Normal	Closed-form transformed MLE	Parameters, original-scale likelihood, CDF, quantiles	Passed
Ln-Normal	SciPy with parameter conversion	Parameters, likelihood, CDF, quantiles	Passed
Exponential	Closed form and SciPy	Parameters, likelihood, CDF, quantiles	Passed
Gamma	SciPy	Parameters, likelihood, CDF, quantiles	Passed
Generalized Extreme Value	SciPy with shape-sign conversion	Parameters, likelihood, CDF, quantiles	Passed
Generalized Logistic	R <code>lmomco</code>	Parameters, likelihood, CDF, quantiles	Passed
Generalized Normal	R <code>lmomco</code>	Parameters, likelihood, CDF, quantiles	Passed
Generalized Pareto	SciPy differential evolution	Global optimum, CDF, quantiles	Passed
Gumbel	SciPy	Parameters, likelihood, CDF, quantiles	Passed
Kappa Four	Analytical zero-shape branch and SciPy finite-shape cases	PDF derivative, CDF-quantile inversion, support	Passed
Logistic	Closed form and SciPy	Parameters, likelihood, CDF, quantiles	Passed
Log-Pearson Type III	Transformed SciPy Pearson III	Parameters, likelihood, CDF, quantiles	Passed
Pearson Type III	SciPy	Parameters, likelihood, CDF, quantiles	Passed
Weibull	SciPy	Parameters, likelihood, CDF, quantiles	Passed

The acceptance tolerances are stored with each oracle. Optimizer coordinates generally use scaled tolerances because BestFit differential evolution and external local optimizers stop by different criteria. Same-point likelihood, CDF, and quantile comparisons retain tighter absolute tolerances.

Log10-Normal closed-form test

For transformed values $x_i = \log_{10}(y_i)$ equal to [1.1, 1.4, 1.7, 2.0, 2.3, 2.6, 2.9] the closed-form MLE is

$$\hat{\mu} = 2, \quad \hat{\sigma} = 0.6.$$

The original-scale likelihood includes $-\sum_i \log(y_i \ln 10)$. The analytical log likelihood is `-44.431208784723104`, the median is `100`, and the 0.9 quantile is `587.3959385303014`.

Quantity	Independent value	BestFit acceptance	Result
$\hat{\mu}$	2.0000000000	absolute error at most <code>1e-5</code>	Passed
$\hat{\sigma}$	0.6000000000	absolute error at most <code>1e-5</code>	Passed
Log likelihood	-44.431208784723104	absolute error at most <code>1e-8</code>	Passed
Median CDF	0.5	absolute error at most <code>1e-12</code>	Passed
0.9 quantile	587.3959385303014	absolute error at most <code>1e-9</code>	Passed

Common-data fitting test

Gumbel, Normal, and Logistic were fitted to the same deterministic sample by BestFit and SciPy. The optimum coordinates and information criteria were compared, and the fitted ranking had to agree exactly.

Gumbel coordinate	SciPy optimum	BestFit	Relative error	Limit	Result
Location	93.11234799935337	93.11293212949110	6.27e-6	1e-4	Passed
Scale	13.157628567998076	13.157823249599968	1.48e-5	1e-4	Passed

Maximum log likelihood, AIC, and BIC agreed under 1e-8 absolute plus 1e-7 relative tolerance. The RMSE equation and inverse-RMSE weights were independently recomputed from BestFit's fitted coordinates at 1e-10 and 1e-12, respectively.

Parameter-adjusted RMSE

For residuals $[1, 2, 3, 4]$ $n = 4$, and $k = 1$, the independent value is

$$\sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2}{4 - 1}} = \sqrt{10} = 3.1622776601683795.$$

BestFit reproduced this value within 1e-12, and paired permutation of observed and fitted values left the statistic unchanged.

Bulletin 17C published examples

The seven official examples exercise systematic records, low outliers, zero-flow years, broken records, perception thresholds, historical intervals, and paleoflood information. Each cell fits the LP3 parent through the specialized GMM path and compares the base-10 log-space mean, standard deviation, and skewness with the published values at absolute tolerance 1e-3.

Example	Principal data condition	Published mean	Published standard deviation	Published skewness	Result
1	68-year systematic record	3.328623159	0.140287994	0.396626124	Passed
2	Low outliers and zero-flow years	3.022663041	0.682087092	-0.929108050	Passed
3	Broken record, thresholds, and low outliers	3.759834285	0.243406211	0.144442997	Passed
4	Historical intervals and perception thresholds	3.885777246	0.245920859	0.817849937	Passed
5	Variable crest-stage thresholds and low outliers	3.278686106	0.233135027	-0.925407257	Passed
6	Historical information and multiple low-outlier detection	3.069106533	0.489820622	-0.462278724	Passed
7	Paleoflood intervals and long perception thresholds	4.653457000	0.376721000	-0.101163000	Passed

Aggregate result: 21 of 21 parameter comparisons passed. Three additional plotting-position cells reproduced PeakFQ results for Examples 4, 5, and 7.

Bootstrap refit reliability

Ordinary and pivotal resampling were applied to each of the seven examples. Examples 1 through 6 used 1,000 refits for each method; Example 7 used 500 for each method, for 13,000 requested outputs overall. The test requires a finite result for every requested output and records retries, substitutions, bound repairs, and numerical clips.

Resampling method	Cells	Requested outputs	Finite outputs	Retries	Parent substitutions	Exceptions
Ordinary bootstrap	7	6,500	6,500	0	0	0
Pivotal bootstrap	7	6,500	6,500	0	0	0
Total	14	13,000	13,000	0	0	0

This result verifies numerical completion and accounting for the published examples. It is not an interval-coverage claim.

Conclusion

All fifteen univariate families passed the declared analytical or external-package comparisons, both multi-candidate fitting cells passed, the seven Bulletin 17C example parameter vectors reproduced the published values, and all fourteen refit-reliability cells completed without substitution or retry. Cohn interval values and broad Bulletin 17C interval coverage remain outside the claims supported by this checkpoint.

CHAPTER 07

Point-Process Analysis

The independent fixture generates Poisson event counts through exponential inter-arrival times and generalized-Pareto marks through an analytical inverse. It separately checks empirical event exposure, fitted threshold intensity, seasonal block assignment, calendar-year versus October-water-year coordinates, and mixed-observation likelihood contributions. Recovery uses 1,000 observations and production DEMCzs defaults.

Verification cell	Evidence	Duration	Result
Calendar-year prior placement	Deterministic occurrence-histogram oracle	0.617 s	Passed
Water-year prior placement	Rotated occurrence-histogram oracle	0.722 s	Passed
Calendar-year seasonal recovery	Generating parent and both changepoints	35.690 s	Passed
Water-year seasonal recovery	Same parent under shifted block origin	33.897 s	Passed
Nonseasonal production recovery	Generating parent	8.450 s	Passed
Seasonal production recovery	Generating parent and both changepoints	36.309 s	Passed
Nonseasonal simulation	Poisson rate and conditional-tail oracle	0.293 s	Passed
Seasonal simulation	Rates, assignment, dates, and conditional tails	0.334 s	Passed
Nonseasonal mixed likelihood	Independent density/CDF calculation	0.294 s	Passed
Seasonal mixed likelihood	Independent annual-maximum calculation	0.291 s	Passed

The paired calendar and water-year fixtures hold changepoints at days 170 and 350 and change only the block-year origin. Generated magnitudes and block days agree exactly; dates differ by 92 days; the parent likelihoods agree within $1\text{e-}10$. The mixed-likelihood comparison uses $2\text{e-}7$ tolerance. All ten cells passed without changing the production likelihood, sampler settings, seed, or acceptance rule.

CHAPTER 08

Competing-Risk Analysis

For a Gaussian copula with latent correlation ρ , the independent oracle uses

$$\rho_S = \frac{6}{\pi} \arcsin(\rho/2), \quad P[\max(X_1, X_2) \leq 0] = \frac{1}{4} + \frac{\arcsin(\rho)}{2\pi}.$$

Each dependence mode generated 40,000 observations with seed 24681357. Empirical rank tolerance was six standard errors, $6/\sqrt{n-1}$; CDF tolerance was six binomial standard errors plus one observation.

Dependence mode	Target	Result
Independent	$\rho_S = 0$, maximum CDF 0.25	Passed
Perfect positive	$\rho_S = 1$, maximum CDF 0.5	Passed
Perfect negative limit	Equation (C.1) at the implemented numerical limit	Passed
User correlation matrix	Equation (C.1) at the configured correlation	Passed

Ten supported one- through three-component minimum/maximum designs were generated with seed 12345. Two-component cases used 1,000 observations and three-component cases 1,500. Every MLE fit passed the declared combined-CDF recovery rule. Four independent-minimum Bayesian designs also passed under production defaults.

Recovery group	Cells	Acceptance	Result
MLE, all supported fixture designs	10	Maximum CDF error 0.05 or 0.06	10 of 10 passed
Bayesian, independently supported minimum designs	4	CDF bound, $\hat{R} < 1.1$. ESS greater than 100	4 of 4 passed

The report makes no Bayesian recovery claim for six difficult maximum or correlated designs whose posterior geometry, convergence, or uncertainty-curve behavior did not satisfy the predeclared criteria. No tolerance, prior, sampler, seed, or likelihood was changed to convert those designs into passes.

CHAPTER 09

Mixture Analysis

BestFit exposes the physical K weights at the model and EM boundary while sampling an identified $K - 1$ vector and deriving the final weight. Tests compare pre-fit data likelihood with Numerics at $1e-10$, fitted engines at $1e-8$, ordered component recovery within 0.1, and Bayesian recovery with 15% relative tolerance plus a 0.15 floor. Bayesian cells also require $\hat{R} < 1.1$ and ESS greater than 100.

Model	Parity/EM result	Bayesian result	Sample size
Two-component Normal mixture	Passed, 1.426 s	Passed	1,000
Zero-inflated two-component Normal mixture	Passed, 1.852 s	Passed	1,000
Three-component Normal mixture	Passed, 3.247 s	Passed	1,000

The three Bayesian runs used production DEMCs defaults and completed in the observed range 173 to 456 seconds. The positive-hurdle cell additionally verifies the point mass at zero and the positive-conditioned continuous components. All six reported parity/recovery cells passed their declared acceptance criteria.

CHAPTER 10

Composite Analysis

Composite analysis combines already estimated child distributions and therefore has no separate fitted parameter vector. Independent tests use direct Normal CDF formulas, published R `mistr` Table 45 quantiles, bivariate-Normal orthants, and complete enumeration of three child posterior supports.

Verification cell	Independent reference	Tolerance	Result
Weighted three-Normal CDF	Analytical weighted sum	$1e-12$ absolute	Passed
Twenty-five mixture quantiles	Published R <code>mistr</code> table	1% relative	Passed
Quantile inversion	Analytical weighted CDF	Probability-scaled, floor $1e-8$	Passed
Independent and comonotonic maximum	Product and minimum CDF identities	$1e-10$ absolute	Passed
Independent and comonotonic minimum	Union and maximum CDF identities	$1e-10$ absolute	Passed
Combination-rule bracketing	Theoretical bounds	$1e-12$ slack	Passed
Mixture posterior	Complete 20 by 20 by 20 Cartesian oracle	Mean 0.02; limits 0.05	Passed
Maximum posterior	Complete Cartesian oracle	Mean 0.02; limits 0.05	Passed
Minimum posterior	Complete Cartesian oracle	Mean 0.02; limits 0.05	Passed
Correlated min/max	Bivariate-Normal orthants at $\rho = 0.6$	$1e-8$ absolute	Passed

Two additional posterior-resampling tests compare the production finite resampling policy with an empirical Cartesian product and a closed-form independent Normal sum. Posterior means pass within 0.02 and interval limits within 0.05, while deliberately raw-paired chains miss by at least 0.10. This demonstrates that each separately fitted source is independently indexed rather than coupled by retained-array position. All twelve reported cells passed.

CHAPTER 11

Bivariate and Coincident-Frequency Analyses

Bivariate copula estimation

An independent Python/SciPy implementation fits six one-parameter copula families to twelve fixed samples of one hundred matched observations. It implements each copula density directly, checks it against a numerical mixed derivative of the CDF, performs a 401-point grid scan, and refines the optimum with bounded minimization. BestFit must reproduce the independent optimum and maximum log likelihood; the historical R `copula` target is retained as a secondary comparison.

Family	Method	Independent optimum	Historical R target	Difference	Result
Ali-Mikhail-Haq	MPL	0.8321521	0.8321504	+1.8e-6	Passed
Ali-Mikhail-Haq	IFM	0.8392378	0.8392506	-1.3e-5	Passed
Clayton	MPL	1.5340162	1.5340160	+1.9e-7	Passed
Clayton	IFM	1.4852343	1.4851670	+6.7e-5	Passed
Frank	MPL	7.7187608	7.7187610	-2.2e-7	Passed
Frank	IFM	8.1303210	8.1302590	+6.2e-5	Passed
Gumbel	MPL	2.0979531	2.0979530	+6.6e-8	Passed
Gumbel	IFM	2.0311985	2.0310340	+1.6e-4	Passed
Joe	MPL	2.6643256	2.6643260	-4.5e-7	Passed
Joe	IFM	2.9656127	2.9652690	+3.4e-4	Passed
Gaussian	MPL	0.8000853	0.8000820	+3.3e-6	Passed
Gaussian	IFM	0.7871334	0.7871479	-1.5e-5	Passed

Production optima agree with the independent coordinates within 1e-5 relative with a 1e-6 scale floor, and maximum log likelihood agrees within 1e-8. Historical R coordinates use a broader 1e-3 compatibility tolerance because those fixtures used a different pseudo-observation convention.

Bivariate recovery and coincident frequency

Seven copula families were generated and recovered with production DEMCz defaults. All seven Bayesian cells passed; the Student t cell completed in 470 seconds. Coincident-frequency response surfaces for independent, positively correlated, and negatively correlated Normal sums were compared with the closed form

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y).$$

All three response-surface cells passed. Independent posterior resampling also matched the closed-form Normal-sum posterior mean within 0.02 and interval limits within 0.05.

Conclusion

Twelve independent copula-optimum cells, seven Bayesian copula-recovery cells, three closed-form coincident-frequency cells, and one posterior-resampling cell passed. The recovery claim remains conditional on the fixed marginal fits, paired events, and parameter regions used by the fixtures.

CHAPTER 12

Rating-Curve Analysis

Discharge-space likelihood

The rating curve is a one-, two-, or three-control piecewise power law. Measurement error is Normal in base-10 log discharge, but the public data likelihood is a discharge-space density. The independent SciPy artifact therefore includes the change of variables for every positive observed discharge.

Model	Independent discharge-space log likelihood at truth	BestFit tolerance	Result
One control	-1577.7594156516989	1e-8	Passed
Two controls	-1850.3134085800252	1e-8	Passed
Three controls	-1888.5687896304166	1e-8	Passed

Scalar, pointwise, and component likelihoods agree with the SciPy density. The Numerics base-10 LogNormal density independently reproduces the same terms.

Activation continuity

At an activation stage h_k , an added control contributes $10^{a_k}(h - h_k)^{\beta_k}$. The analytical cells evaluate offsets from 1e-3 through 1e-9 and exponents 0.1, 1, 1.67, and 2.5. The two-sided increment agrees with the power-law formula within 1e-10 relative tolerance. With the default lower bound $\beta = 0.1$, the increment ratio between offsets 1e-12 and 1e-6 is

$$(10^{-12}/10^{-6})^{0.1} = 0.2511886432.$$

The contribution therefore vanishes continuously at activation. All seven continuity and bound cells passed.

Recovery

The publication recovery fixture uses 1,000 observations generated from the same source-project parameterization as the worked examples. The independent optimum uses SciPy multistart least squares on base-10 residuals with a profiled scale.

Model	Generating parameters	Independent MLE parameters	Result
One control	[1, 0.243897, 2.666667, 0.05]	[1.000633, 0.244804, 2.664527, 0.052251]	Passed
Two controls	[1, 0.243897, 2.666667, 10, 2.946300, 1.67, 0.05]	[1.000989, 0.246160, 2.661335, 9.981416, 2.936743, 1.678032, 0.052240]	Passed
Three controls	[1, 0.243897, 2.666667, 10, 2.946300, 1.67, 15, 3.60953, 3, 1.67, 0.05]	[1.001032, 0.246300, 2.660905, 9.936705, 2.899111, 1.730227, 15.332091, 3.828382, 1.341175, 0.051935]	Passed

MLE acceptance requires same-point likelihood agreement, optimizer optimality, coordinate tolerances, curve parity within 0.5%, and the fitted curve within 10% of the generating curve. Bayesian acceptance requires $\hat{R} < 1.1$, ESS greater than 100, sampled-MAP parity, MAP-curve parity within 2%, and coverage of the generating curve by the central 90% band.

Bayesian model	Grid stages containing truth	Grid stages tested	Result
One control	36	36	Passed
Two controls	36	36	Passed
Three controls	34	36	Passed

Twenty additional synthetic recovery cells, ten MLE and ten Bayesian, used 1,000 observations and production defaults; all passed.

Conclusion

Rating-curve discharge-space likelihood, activation continuity, example replication, and the twenty supporting recovery cells passed all 36 reported cells. The evidence does not establish parity for arbitrary control matrices or uncertainty coverage in extrapolation.

CHAPTER 13

Time-Series Analyses

AR, MA, ARIMA, and ARIMAX oracle design

The fixtures were generated independently in R without calling BestFit or Numerics. Each fixture discards 110 initialization steps and retains 1,000 observations. The deterministic oracle set independently reconstructs scale priors, invalid-scale behavior, transform fitting, conditional likelihood alignment, prediction recurrence, prediction variance, transformed generation, and information criteria.

Oracle group	Independent calculation	Tolerance	Result
Jeffreys scale metadata	Four scalar scale-prior evaluations	1e-12	Passed
Invalid scale	Scalar and pointwise Gaussian likelihood and prior	1e-12; exact rejection	Passed
Transform parameter	R training-only Box-Cox/Yeo-Johnson profiles	Stored cross-language tolerance	Passed
Manual transform rebuild	Independently transformed likelihood	Stored cross-language tolerance	Passed
ARIMAX alignment	Date-indexed differencing, covariates, lags, and Jacobian	1e-10	Passed
ARIMA/ARIMAX reintegration	Hand recurrence for irregular holdout boundaries	1e-10	Passed
Forecast uncertainty	Boundary-conditioned variance recurrence	1e-10	Passed
Transformed forecasts	Model-scale recurrence followed by one inverse transform	1e-10	Passed
AR/MA transformed generation	Algebraic and 1,000-step moment oracles	Declared moment bounds	Passed
ARIMA transformed generation	Independent transformed/differenced recurrence	Declared moment bounds	Passed
ARIMAX transformed generation	Independent date/covariate recurrence	Declared moment bounds	Passed
Information criteria	Data likelihood at stored MAP, prior excluded	1e-10	Passed

The ARIMA forecast oracle anchors the first prediction to the final observed transformed level and keeps AR and MA recurrences on the model scale until one inverse response transform. For the declared ARIMA(1,1,1) fixture, the next zero-innovation difference is -0.0290494944510721 and the next original-scale response is 332.259097907995 . For the ARIMAX fixture the corresponding values are -1.2555476231495 and 251.211255651498 .

Recovery

Model and estimator	Generating coordinates	Sample size	Acceptance	Result
AR(1), MLE	$\mu = 10, \phi = 0.6, \sigma = 5$	1,000	Existing coefficient and scale gates	Passed
AR(1), Bayesian	Same	1,000	Central interval, $\hat{R} < 1.1$, ESS greater than 100	Passed
MA(1), MLE	$\mu = 10, \theta = 0.6, \sigma = 5$	1,000	5% parameter gate	Passed
MA(1), Bayesian	Same	1,000	Central interval, $\hat{R} < 1.1$, ESS greater than 100	Passed
ARIMA(1,1,1), MLE	log transform; $\phi = 0.45, \theta = 0.25, \sigma = 0.04$	1,000	Independent conditional optimum and forecast oracle	Passed
ARIMA(1,1,1), Bayesian	Same	1,000	Independent posterior MAP, 95% interval, diagnostics	Passed
ARIMAX(1,1,0), MLE	intercept 0.25, $\beta = 1.5, \phi = 0.4, \sigma = 0.5$	1,000	Date-indexed independent optimum	Passed
ARIMAX(1,1,0), Bayesian	Same	1,000	Independent posterior MAP, 95% interval, diagnostics	Passed

An additional matrix covers 29 AR, MA, ARIMA, and ARIMAX configurations with 1,000 observations. Seven MLE cells retained their parameter tolerances. Twenty-two Bayesian cells require every generating value inside the central 90% interval and $\hat{R} < 1.1$. All 29 passed. The independent eight-cell matrix above remains the primary oracle-backed recovery evidence.

Conclusion

Transform, likelihood, recurrence, forecast-boundary, uncertainty, generation, and information-criterion calculations passed twelve independent oracle groups. The eight principal recovery cells and 29 supporting recovery cells passed within their declared scope.

CHAPTER 14

Spatial-Extremes Analysis

Likelihood oracle

The fixture has five sites, twelve row/year blocks, and structured missing observations. An R `mvtnorm` generator evaluates Hosking-sign GEV margins, the Gaussian copula on the observed-site correlation submatrix, latent Gaussian-process densities, and finite-difference gradients.

Row	Observed sites	Independent marginalized log likelihood
5	1, 3, 4, 5	-23.7536861453931
6	2, 3, 4	-14.894701
7	1, 2, 4, 5	-21.970089
8	1, 5	-9.952734
9	None	0

The full marginalized data log likelihood is -238.89272936973 . BestFit agrees within $1e-8$; every row agrees within $1e-10$ and sums to the scalar result.

Spatial likelihood comparison	Acceptance	Result
Missing-site scalar marginalization	$1e-8$	Passed
Missing-site pointwise rows	$1e-10$	Passed
Complete-row copula likelihood	Stored oracle tolerance	Passed
Marginal-only GEV likelihood	Stored oracle tolerance	Passed
Posterior-kernel decomposition	Exact data plus prior identity	Passed
Data likelihood excludes process density	Stored oracle tolerance	Passed
Scalar and pointwise decomposition	Stored oracle tolerance	Passed
Scalar and pointwise gradients	$1e-4$	Passed

The criteria cell used eleven nonempty row/year blocks. AIC and BIC were recomputed from the observation likelihood at the sampled MAP within $1e-8$; WAIC and its effective-parameter term agreed with a row/year recomputation within $1e-9$ relative; one Pareto value was produced per row/year.

Prediction, distance, uncertainty, and recovery

Verification area	Independent reference or design	Result
Conditional prediction	R Gaussian-process conditional mean and variance, 15 cases, $1e-10$	Passed
Geodesic distance	R haversine distance, correlation, and kriging moments	Passed
Cartesian distance	Exact planar Euclidean distances	Passed
Leave-one-site-out	Independently reduced models with held-out covariates	3 of 3 passed
Ungauged prediction	Posterior mean of per-draw conditional GP prediction	Passed
Regional curve	Posterior of the per-draw regional mean quantile	Passed
Dependent simulation	20,000 rows; correlations within 0.02; quantiles within 3%	Passed
Bayesian interval inflation	Width increased by the square root of the variance inflation factor	Passed
Godambe path	Explicit unavailable status for singular sensitivity; finite draws when available	Passed
Temporal block bootstrap	20 fitted replicates with complete accounting	Passed

Complete-data recovery added two MLE and seven Bayesian cells under production defaults. All nine passed. Bayesian durations ranged from 42 to 115 seconds.

Conclusion

Spatial likelihood, missing-data marginalization, kriging, distance, criteria, cross-validation, prediction, regional uncertainty, dependent simulation, method dispatch, and nine recovery cells passed all 30 reported cells. The evidence remains conditional on the tested network sizes, covariates, missingness patterns, and correlation structures.

CHAPTER 15

Evidence Boundaries and Conclusions

Supported conclusions

The report supports the following conclusions for the documented software checkpoint:

- The fifteen univariate distributions reproduce their declared analytical, SciPy, or R package references over the tested parameter regions.
- Fitting-analysis criteria, ranking, and model weights reproduce independent calculations for the declared common-data fixture.
- MLE, MAP, GMM, DIC, WAIC, PSIS-LOO, profile, covariance, and rank-normalized MCMC diagnostic calculations reproduce analytical or R package references in the tested cases.
- The specialized Bulletin 17C GMM path reproduces the seven published example parameter vectors and completes the declared bootstrap reliability matrix.
- Point-process, supported competing-risk, mixture, and composite calculations reproduce analytical identities or recover their declared generating designs.
- Bivariate copula estimation and coincident-frequency response surfaces reproduce the declared independent references.
- Rating curves reproduce independent discharge-space likelihoods, analytical continuity, and generating curves.
- Time-series models reproduce independent transform, likelihood, recurrence, prediction, uncertainty, generation, and recovery results.
- Spatial likelihoods, prediction, distance, uncertainty, and recovery reproduce the declared independent references.

Claims not established by this report

The following boundaries prevent readers from extending the conclusions beyond the evidence:

- **Bulletin 17C Cohn values.** Exact-data LP3 scope guards are tested, but numerical Cohn interval values are not verified at this checkpoint.
- **Bulletin 17C coverage.** Previously defined coverage cells were not executed for this checkpoint. Bootstrap reliability is not a substitute for interval coverage.
- **Competing-risk Bayesian generality.** Six maximum or correlated designs are excluded from the supported Bayesian matrix because their convergence, effective sample size, identifiability, or uncertainty-curve criteria were not satisfied. Their MLE cells remain supported.
- **Joint bivariate estimation.** Copula estimation and recovery condition on fitted marginals. A joint marginal-copula posterior is not implemented or verified.
- **Generic prior-predictive sampling.** Independently sampled parameter priors do not reproduce coupled quantile, Jeffreys, spatial-error, or other joint prior factors.
- **Spatial scale.** The tested networks are modest. Computational performance, approximation quality, and calibration for large networks are not established.
- **Spatial weighting.** Correlation heuristic weights change relative site influence. They are not a pairwise composite likelihood, an effective sample size, or a Godambe correction.
- **PSIS scope.** BestFit reports Pareto diagnostics but does not perform exact leave-one-out refits, moment matching, or chain-relative-efficiency adjustment.
- **Application validation.** Passing synthetic and published-example cells does not establish that a selected model is physically adequate for a specific watershed or decision problem.

Software-quality controls

Fast regression projects protect constructor validation, state transitions, exceptions, deterministic calculations, caching, public API snapshots, and serialization. At the publication checkpoint the recorded regression gate contained 3,342 core, 579 UI, 438 App, and 498 API tests with zero failures, and the strict XML documentation gate passed. These results support implementation stability but are not counted as independent numerical verification.

Reproducibility

Each independent oracle records its generator, package versions, source inputs, parameter conversions, seed, and hash. The data manifest covers 29 committed artifacts. Long-running verification methods are invoked one at a time by fully qualified name, and each invocation must produce one TRX result. The public report is generated from an explicit book manifest; report metadata identifies the exact BestFit and Numerics source commits.

Overall conclusion

RMC.BestFit 2.0 has broad, multi-source numerical evidence across its principal statistical and hydrologic capabilities. Exact analytical and external-package comparisons establish the calculation-level core; published examples establish practical parameter parity; recovery tests exercise estimators and posterior propagation; and the limited coverage studies establish only their declared simulation claims. All results presented as passing in this report satisfied predeclared acceptance rules. The evidence boundaries above are intentionally excluded from the verified claim set and should guide external review, future verification priorities, and application-specific model validation.

CHAPTER 16

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